KubeVision

An Image Classification Pipeline on Kubernetes — Milestone 1: Architecture

Course: CSC 478 — Cloud Engineering

Instructor: Dr. Linh Ngo

Student: Angeudy Brisita Perez

Milestone: 1 of 3 — Architecture (design only)

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Repository: https://github.com/angeudybp/csc478-project

Project idea: Image Classification Pipeline on Kubernetes

1. Overview

Photos pile up faster than anyone can sort them. Tagging a folder of pictures, checking what a trail camera caught, or routing product photos to the right catalog section all start with the same question: what is in this image? KubeVision answers it. Upload a JPEG or PNG and get back the most likely label, how confident the model is, and the next two candidates, all in one request. Every upload is saved with its result, so a past prediction can be looked up later and both the original and the exact 224×224 image the model saw can be downloaded. Behind that single request is a small pipeline built to keep working: each stage runs as its own Kubernetes service, the model tier scales out to three replicas, and a failed pod is replaced while the others keep serving. KubeVision follows the course's template project, an image classification pipeline on Kubernetes, and uses a small pre-trained model (MobileNetV2) so the work goes into orchestration rather than training.

KubeVision is a small image classification service on Kubernetes. A user uploads an image to a single REST endpoint and gets back the label and the confidence score. Four microservices perform different parts of the task:

- **Gateway** — performs uploads validation and orchestrates the rest of the services;
- **Preprocess** — resizes and normalizes images for the model;
- **Inference** — executes MobileNetV2 model (scaled to 3 replicas with metrics sidecar);
- **Storage** — stores the original image, the preprocessed image, and the output so that they could be retrieved later.

Everything runs on a three-node RKE2 cluster (one server, two agents) on the FABRIC testbed, built with the course's Ansible playbooks. The same manifests also run unchanged on single-node Minikube, which is used for local development and testing. Three design choices drive the architecture:

1. **A centralized gateway** instead of chain of calls from one service to another. The solution introduces one additional network hop, but ensures that there is precisely one public entrypoint in the system and all the other components are easy to implement.
2. **Synchronous requests** instead of message queue. While the queue could handle a higher load more efficiently, implementing it would mean moving the project focus from Kubernetes to distributed messaging.
3. **Single service to store the data.** Only the storage service has a persistent volume.

2. User Stories and Data Flow

US-1 — Image classification. *I want to send an image, to get the predicted class of it with the corresponding confidence score so that I can use it for image classification without installing any software.*

• POST /api/v1/images with a valid file less than 8MB returns 200 with the imageId, label, confidence and the first 3 classes.

• *The image file bigger than 8 MB results in error 413; non-image file bigger than 0 bytes results in 415. None of the errors causes pod crash.*

• *Tests: T1-T6.*

US-2 — Retrieve a result. *I want to be able to get my previous prediction and the associated with it image IDs to have my results available for retrieval later.*

• *GET /api/v1/results/{imageId} gives me the saved prediction and if I request a wrong imageId, the system returns 404 error.*

• *I can download the original and the processed images using image ID.*

• *The results stay available even after recreating the storage pod. Tests: T7, T11.*

US-3 — Scale out and survive. *I want to scale the inference tier and lose individual pods, so that the traffic never stops.*

• *The inference scales up to 3 replicas and the metrics sidecar shows the request load on each one of the pods.*

• *Deleting one of the inference pods automatically triggers creating it while the rest continue working.*

• *The gateway returns 503 code, when I stop the inference, instead of hang.*

• *kubectl delete -f . && kubectl apply -f . restores the whole stack without additional actions. Tests: T8-T10, T12, T13.*

Figure 1 depicts the data path. An image upload goes through the pre-processing, inference and storage stages. Retrieving involves only gateway and storage services.

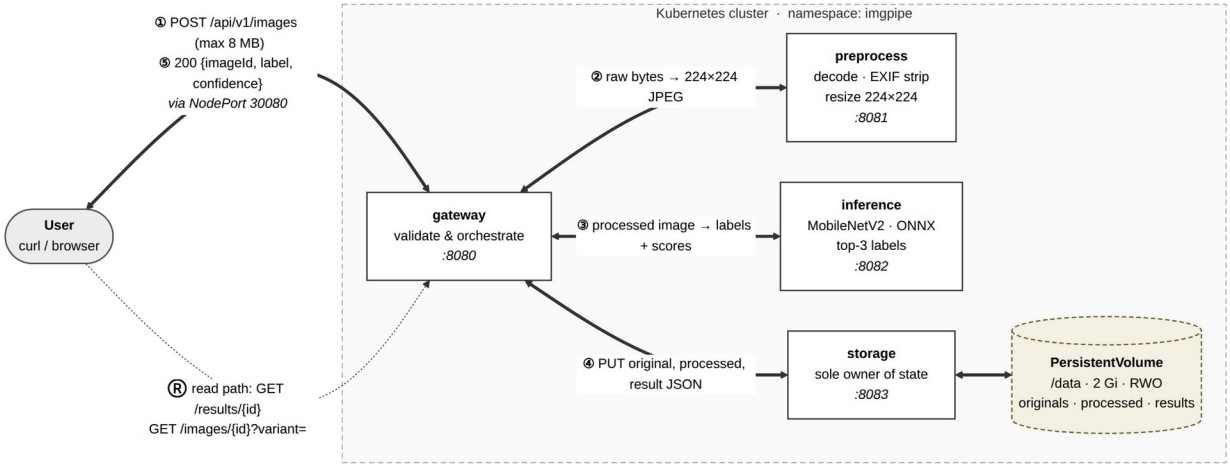


Figure 1. Data flow. Solid arrows are the upload path; dashed arrows are the retrieval path. Only the gateway is reachable from outside the cluster.

3. Architecture

The design follows the brief's constraints: exactly one external entrypoint, at least one multi-container pod, and one component scaled to three or more replicas. It targets a three-node RKE2 cluster on FABRIC, which has no cloud load balancer and keeps persistent data on an NFS share exported by the server node. Minikube runs the same manifests for local testing.

3.1 Components

Component	Containers	Replicas	Port	Responsibility
`gateway`	1	1	8080	Only public entrypoint. Validates uploads, calls the other services in order, and serves retrieval requests.
`preprocess`	1	2	8081	Decodes the image, removes metadata, resizes to 224×224, and normalizes it (Pillow).
`inference`	**2**	**3**	8082, 9100	Runs MobileNetV2 with ONNX Runtime and returns the top labels. Includes the metrics sidecar.
`storage`	1	1	8083	Owns the only persistent volume. Saves images and prediction records.

Table 1. Components. Inference is the only multi-container pod.

Why a separate gateway? Without it, the preprocessor would have to coordinate the whole pipeline, and storage would need to be public for retrieval to work. A gateway keeps preprocessing and inference focused on one job each and keeps stored data off the public network.

Why is preprocessing its own service? It is the stage most likely to change (image size, model type), and it is CPU-heavy while the gateway mostly waits on network calls. Splitting them lets each be scaled or replaced on its own.

3.2 Multi-Container Pod: Inference + Metrics Sidecar

Each inference pod includes two containers using an emptyDir volume at /var/log/app/:

- **inference** — model server, writes one JSON line (time, request ID, latency, result) per request into requests.jsonl.
- **metrics-exporter** — sidecar, reads the file and provides the request count, error count and p50/p95 latency statistics via HTTP at 9100 port.

They must be in one pod because the sidecar must read the file written by the application and must provide statistics for the specific replica. Separate Deployments cannot use shared emptyDir volumes. The placement of the sidecar on the scaled component proves the effectiveness of the load balancing – if there are three replicas, their metrics endpoints must have about the same number of requests. Drawback – metrics are stored in memory and are lost after the restart, and no Prometheus is provided; metrics are gathered via kubectl port-forward and curl.

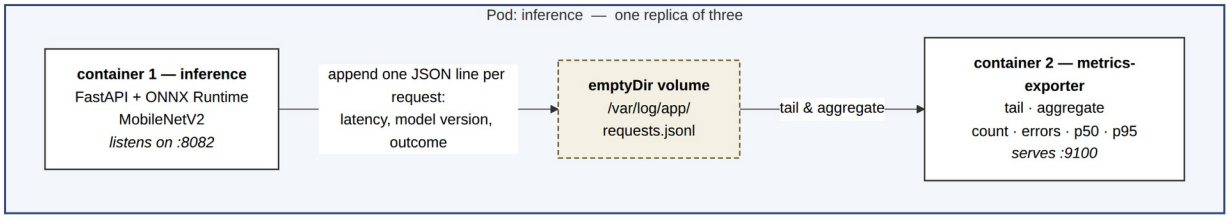


Figure 2. The inference pod. Both containers share a volume and network, so they run in one pod. Each replica has its own exporter.

3.3 Services and External Access

Service	Type	Ports	Reason
`gateway-svc`	**NodePort**	30080 → 8080	The single public entrypoint. Neither RKE2 on FABRIC nor Minikube has a cloud load balancer, so a `LoadBalancer` Service would stay `Pending`; NodePort works on both. On FABRIC it is reached through an SSH tunnel to a cluster node.
`preprocess-svc`	ClusterIP	8081	Internal only; balances across the two replicas.

Service	Type	Ports	Reason
`inference-svc`	ClusterIP	8082, 9100	Internal only; balances across the three replicas.
`storage-svc`	ClusterIP	8083	Internal only; only the gateway calls it.

Table 2. Service choices. Only `gateway-svc` is reachable from outside the cluster.

Services are reached by DNS name (e.g. `preprocess-svc.imgpipe.svc.cluster.local`), so no configuration changes when pods move. Figure 3 shows the full layout.

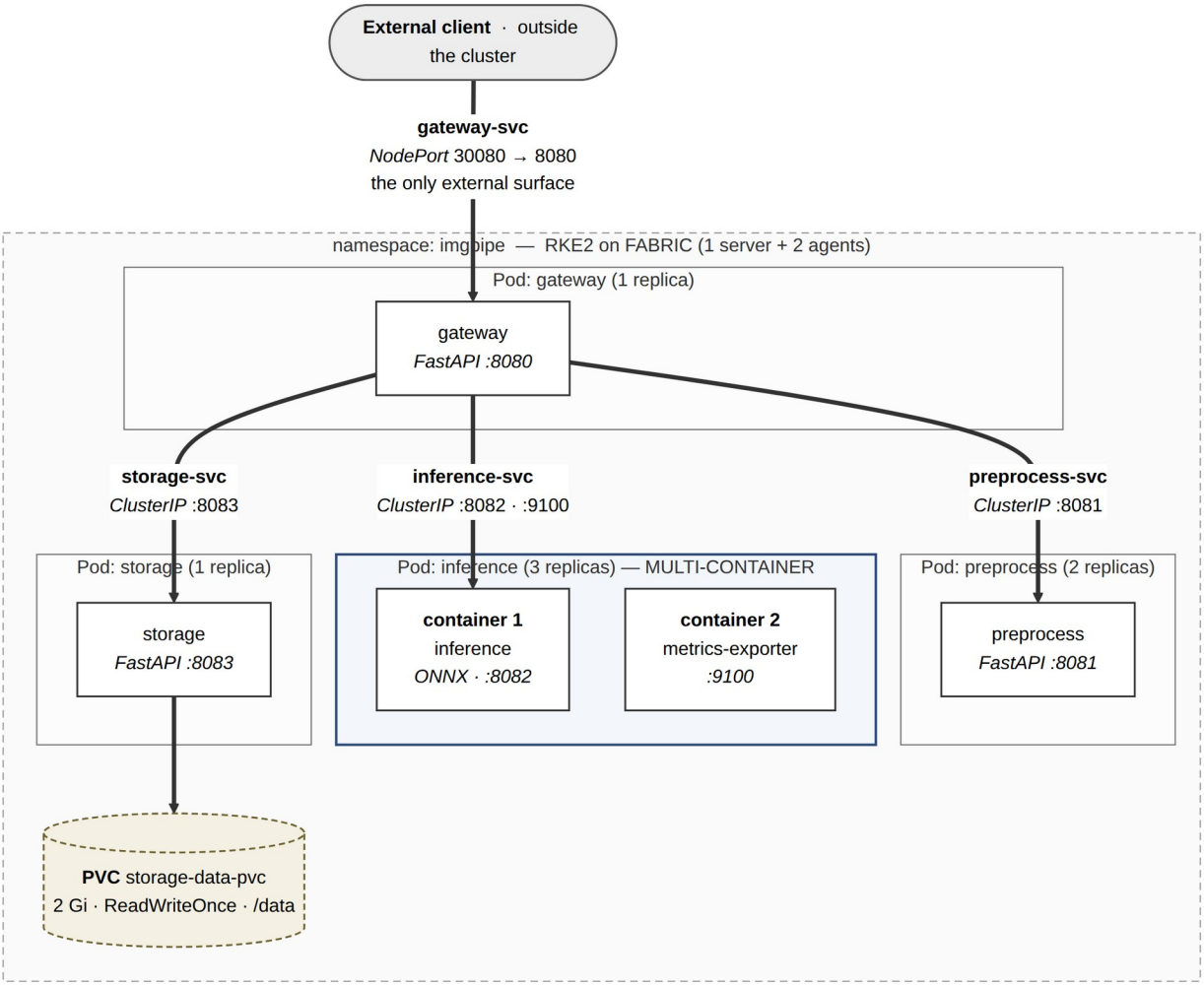


Figure 3. Kubernetes topology. Outlined regions are pods with replica counts; only `gateway-svc` crosses the cluster edge.

3.4 Request Flow and Timeouts

The gateway assigns each request an `X-Request-Id` and passes it to every service, so one request can be traced through all logs. The steps are:

1. **Validate** — check file type and the 8 MB size limit before reading the whole upload.
2. **Preprocess** — send the image to `preprocess` and get back a 224×224 JPEG.
3. **Infer** — send the processed image to `inference` and get back the top labels.
4. **Store** — save the original, processed image, and prediction to `storage`, then respond.

Hop	Timeout (target)	Retry
Client → gateway	30 s	None — the client decides.
Gateway → preprocess	5 s	Once, only if the connection failed.
Gateway → inference	15 s	**Never.** Retrying a slow, overloaded tier makes it worse.
Gateway → storage	5 s	Once, only if the connection failed.

Table 3. Timeout budget. Internal timeouts add up to less than 30 s, so a slow service produces a clean error instead of a dropped connection. These are targets, not measurements.

3.5 Scaling Target

Inference is scaled to 3 replicas. It holds no state, so replicas are interchangeable, and running the model is the most CPU-heavy step in the pipeline, so it will be the first to slow down under load. The gateway stays at 1 replica (it mostly waits on network calls), preprocess runs 2 for availability, and storage is fixed at 1 because its volume can only be attached to one pod (§5).

3.6 Main Tradeoff: Synchronous Requests

The gateway keeps the client connection open for the whole pipeline and returns the prediction in the same response. The alternative is a queue: accept the upload, return a job ID, and let workers process it in the background. A queue handles traffic spikes better, but it adds a message broker, retry handling, and job tracking — a lot of work that is not about Kubernetes. The synchronous design is simpler, and its failures are easy to see. The cost: **when inference is overloaded, requests time out instead of waiting in line.** Adding a queue is planned as a future improvement.

3.7 Reliability Settings

- **Health checks.** Every service has `/healthz` (is the process alive?) and `/readyz` (can it handle traffic?). Inference only reports ready after the model has loaded, so new pods never receive requests too early.
- **Resource limits.** Inference requests 256 Mi memory / 0.25 CPU, limited to 768 Mi / 1 CPU, with `OMP_NUM_THREADS=1` so three replicas do not fight over CPU. Other services: 64 Mi / 0.05 CPU, limited to 256 Mi / 0.5 CPU.
- **Images.** Fixed version tags (never `latest`), published as public packages on GitHub Container Registry so FABRIC nodes can pull them without credentials. For local runs they are copied into Minikube with `minikube image load`. Model weights are built into the image so pods start without downloading anything.

3.8 Repository Layout

Everything runs in the `imgpipe` namespace. Manifest files are numbered so `kubectl apply -f .` creates them in the right order:

```
deployments/   00-namespace.yaml  10-gateway.yaml  11-preprocess.yaml
               12-inference.yaml  13-storage.yaml
services/      20-gateway-svc.yaml  21-preprocess-svc.yaml
               22-inference-svc.yaml  23-storage-svc.yaml
storage/       05-storage-pvc.yaml
services-src/  gateway/  preprocess/  inference/  metrics-sidecar/  storage/
```

4. API Contracts

All services use JSON over HTTP (FastAPI, Python 3.11). Every error uses the same format so the gateway can pass errors along consistently:

```
{ "error": { "code": "PAYLOAD_TOO_LARGE", "message": "Upload exceeds 8 MB" },
  "requestId": "7f3c1e9a" }
```

4.1 Public API (gateway)

Endpoint	Purpose
`POST /api/v1/images`	Upload an image (multipart, JPEG/PNG, max 8 MB) and get a prediction.
`GET /api/v1/results/{imageId}`	Get a saved prediction.
`GET /api/v1/images/{imageId}?variant=original|processed`	Download the original or processed image.
`GET /api/v1/results?limit=20`	List recent predictions, newest first.

Table 4. Gateway endpoints.

Example request and response (through an SSH tunnel that forwards localhost:30080 to the gateway NodePort on FABRIC):

```
curl -X POST "http://localhost:30080/api/v1/images" -F "file=@dog.jpg"

{ "imageId": "img_01JAR7K2M9QX4T8B",
  "label": "golden retriever", "confidence": 0.8734,
  "topK": [ {"label": "golden retriever", "score": 0.8734},
            {"label": "Labrador retriever", "score": 0.0612},
            {"label": "kuvasz", "score": 0.0188} ],
  "modelVersion": "mobilenetv2-1.0-onnx", "mock": false,
  "timings": {"preprocessMs": 38, "inferenceMs": 121, "storageMs": 44, "totalMs": 207},
  "requestId": "7f3c1e9a" }
```

Timings above are example values only.

Code	Error code	When
200	—	Success.
400	`EMPTY_UPLOAD`	Empty file or missing `file` field.
404	`NOT_FOUND`	Unknown `imageId`.
413	`PAYLOAD_TOO_LARGE`	Upload over 8 MB.
415	`UNSUPPORTED_MEDIA_TYPE`	Not a JPEG/PNG, or not a valid image.
502	`UPSTREAM_FAILED`	An internal service returned an error.
503	`UPSTREAM_UNAVAILABLE`	No healthy replica of an internal service.
504	`UPSTREAM_TIMEOUT`	An internal service took too long (Table 3).

Table 5. Gateway status codes.

4.2 Internal APIs

Service	Endpoint	Input → Output
preprocess	`POST /preprocess`	Raw image bytes → 224×224 JPEG (base64) with original dimensions. 415 if not an image.
inference	`POST /predict`	`{ data, topK }` → top labels and scores, model version, `mock` flag.
inference sidecar	`GET /metrics` (port 9100)	Request count, error count, p50/p95 latency (Prometheus format).
storage	`PUT` / `GET /objects/{imageId}/{variant}`	Save or fetch image bytes (`original` or `processed`).
storage	`PUT` / `GET /results/{imageId}`	Save or fetch a prediction record. Saving the same ID twice overwrites it.

Service	Endpoint	Input → Output
storage	`GET /results?limit=n`	List recent records.
all	`GET /healthz`, `GET /readyz`	Liveness and readiness checks (§3.7).

Table 6. Internal endpoints. None are reachable from outside the cluster.

Mock mode. With `MOCK_MODE=1`, inference returns a predictable fake label based on the image bytes, so the pipeline can be built and tested without the real model. Every response includes a `mock` field so fake results are never mistaken for real ones.

5. Persistence Plan

Milestone 2 uses temporary storage; Milestone 3 adds the persistent volume described here. Each upload produces three saved items:

Item	Size (est.)	Why it is saved
Original image	0.1–8 MB	Lets a prediction be re-run later without re-uploading.
Processed image (224×224)	~20 KB	Shows exactly what the model saw.
Prediction record (JSON)	~1 KB	The result: labels, scores, model version, timings, timestamp.

Table 7. Data saved per upload. Logs are not saved to disk; they are read with `kubectl logs`.

Volume. One PersistentVolumeClaim, `storage-data-pvc`: 2 Gi, ReadWriteOnce. On FABRIC it binds to a statically created NFS PersistentVolume on the share that the server node exports to the other nodes over the private network (`/opt/data`, set up by the course's NFS playbook); on Minikube it uses the default `standard` storage class. It is mounted at `/data` in the storage pod only. Reclaim policy is `Retain`, so deleting the claim does not delete the data. Because the data sits on NFS, the storage pod can be rescheduled on any node and still find it. A manually created `hostPath` volume is the backup plan if the NFS share is unavailable.

```
/data/
├── originals/<yyyy>/<mm>/<dd>/<imageId>.<ext>   original upload
├── processed/<yyyy>/<mm>/<dd>/<imageId>.jpg     224x224 model input
├── results/<imageId>.json                       prediction record
└── index/manifest.jsonl                         one line per upload (for listing)
```

Images are grouped by date so no folder grows too large. The index file is append-only, which makes listing recent results fast and avoids rewriting a shared file.

Single-writer limit. A ReadWriteOnce volume can only be attached to one pod, so storage runs **1 replica** with the `Recreate` update strategy (a rolling update would get stuck trying to attach the volume to a second pod). The tradeoff is a short write outage during storage restarts; uploads in that window get a 503. In return, there is never more than one process writing files, so no locking is needed. Moving to an object store such as MinIO would remove this limit and is a possible next step.

Verification (Milestone 3):

1. Upload three images and record their IDs.
2. Confirm all three can be retrieved.
3. Delete the storage pod: `kubectl delete pod -l app.kubernetes.io/component=storage -n imgpipe`.
4. Wait for Kubernetes to recreate it, then retrieve all three again — the results must match exactly.
5. Repeat with `kubectl rollout restart deployment/storage` and with a full `kubectl delete -f . && kubectl apply -f .`, recording whether data survives each.

6. Risk and Test Plan

#	Risk	L / I	Mitigation	Test
R1	**Large model image** slows pod startup or fails to load on the cluster nodes.	M / M	Small model (MobileNetV2, ~14 MB) on a slim base image; weights built in; `imagePullPolicy: IfNotPresent`.	T12
R2	**Timeouts cascade**: one slow service hangs the whole request.	M / H	Per-hop timeouts that add up to less than the client timeout; no retries on errors.	T8
R3	**Out of memory** when inference replicas share a node.	M / H	Memory/CPU limits and `OMP_NUM_THREADS=1`.	T10, T13
R4	**Storage volume** cannot attach to two pods.	L / H	Storage fixed at 1 replica with `Recreate`.	T11
R5	**Requests reach a pod before the model loads.**	H / M	Separate readiness check that waits for the model.	T9, T14
R6	**Huge or malicious images** exhaust memory.	M / H	8 MB limit checked early, file-type allowlist, pixel-count limit in Pillow.	T2, T4, T5
R7	**Teardown/rebuild fails** for grading.	M / H	Numbered manifests, dedicated namespace, fixed NodePort.	T12
R8	**Solo schedule**: getting stuck on the model blocks everything.	M / M	Mock mode lets the pipeline be built without the real model.	T1

Table 8. Top risks. L = likelihood, I = impact (Low / Medium / High).

Test approach. Unit tests (`pytest`) cover each service's logic. Each service is then tested alone with `curl`. Integration tests run the full pipeline through the public endpoint using a script in the repository. Operational tests (scaling, pod deletion, persistence) use `kubectl`, and their output is saved as evidence. Milestone 2 uses mock mode so that Kubernetes problems are not confused with model problems.

ID	Case	Action	Expected result	When
T1	Happy path	Upload a 500 KB dog JPEG	200 with a label; three items saved	M2
T2	Too large	Upload a 12 MB JPEG	413; no pod restart	M2
T3	Near limit	Upload a 7.5 MB JPEG	200 within 30 s	M2
T4	Invalid image	Text file renamed `.jpg`	415; no crash	M2
T5	Wrong format	Upload GIF and SVG	415 for both	M2
T6	Empty file	Upload a 0-byte file	400	M2
T7	Unknown ID	`GET /results/img_doesnotexist`	404	M2
T8	Service down	Scale inference to 0	Quick 503, recovers when scaled back	M2
T9	Self-healing	Delete one inference pod	Pod recreated; other requests keep working	M2
T10	Scaling	Scale inference to 3	All 3 ready; sidecars show balanced requests	M2
T11	Persistence	Upload 3, delete storage pod, re-fetch	Identical results after restart	M3
T12	Clean rebuild	`kubectl delete -f . && kubectl apply -f .`	Everything ready, no manual steps	M2
T13	Concurrency	10 uploads at once	All 200; first measured p95 latency	M2
T14	Readiness	Restart inference, poll `/readyz`	Not ready until model loads; no early traffic	M2

Table 9. Test matrix. M3 tests need the persistent volume.

7. Team Charter

This is a solo project, so the charter defines roles, a working schedule, and a Definition of Done for one person.

Role	Responsibility	Effort
Platform	FABRIC slice and RKE2 cluster, Minikube for local testing, manifests, Services, health checks, resource limits, PV/PVC, teardown and rebuild.	~45%
Services	The four FastAPI apps, the metrics sidecar, Dockerfiles, the model and mock mode.	~35%
Docs & demo	Reports, diagrams, evidence screenshots, demo video.	~20%

Table 10. Roles. Platform gets the most time because the project is graded on Kubernetes work.

Checkpoint	How often	Output
Work session	2–3 per week, ~2 hours	Commits pushed; nothing left uncommitted.
Self-review	Weekly	Short note in `docs/journal.md`: progress, updated risks, blockers.
Milestone check	Before each submission	Go through every deliverable on the checklist.

Table 11. Working schedule.

Definition of Done. A task is done only when all of these are true:

1. It deploys on a fresh cluster with no manual steps.
2. It works when tested with a `curl` command saved in the repository.
3. Its test from Table 9 passes, and the output is saved.
4. `kubectl delete -f . && kubectl apply -f .` still works.
5. Any change from this design is written down with the reason.
6. It is committed with a clear message.

Milestone	Scope
M1	Architecture and design (this document).
M2	Deployments, Services, sidecar, scaling and self-healing evidence; no persistence.
M3	PV/PVC, persistence checks, health checks, resource limits, demo video, final report.

Table 12. Milestones. In M2, the mock pipeline is built first so a working end-to-end system exists early.